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Physics Procedia 39 (2012) 526 – 533

Physics Procedia

LANE 2012

Simple evaluation procedure of nonlinear absorptivity in glass welding using USLP with high-pulse repetition rates

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Abstract

A simple non-dimensional equation has been derived to evaluate nonlinear absorptivity in simplified non-dimensional form, if the cross-sectional area of the molten region is experimentally given in internal melting of glass by ultrashort laser pulses at high pulse repetition rates. Based on a series of experiment using different glasses including D263, Foturan and fused silica in wide variety of experimental conditions, it is shown that the evaluated nonlinear absorptivity has an excellent correlation with the results obtained by previously reported simulation model.

Keywords: nonlinear absorptivity; glass; ultrashort laser pulse; welding; thermal conduction; isothermal line; non-dimensional equation

1. Introduction

Internal modification is one of the most interesting topics in the applications of ultrashort laser pulses (USLP) due to unique laser-matter interaction based on nonlinear absorption process [1,2]. Among wide variety of applications of USLP [3-5], glass welding by USLP has been recently attracting much attention due to advantages of crack-free welding without pre- and post-heating providing selective local joining at the glass interface [6-14], and is applicable to glass independently of coefficient of thermal expansion (CTE) in contrast to the case of existing CO_2 laser welding [15-17] where crack-free welding is limited to glass with low CTE.

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 doi:10.1016/j.phpro.2012.10.069

In internal melting of glass, nonlinear absorptivity is one of the key issues for understanding laser-matter interaction and optimizing welding process. The laser energy absorbed in the laser-induced plasma has been also simulated based on the rate equation for free electrons at single pulse irradiation in distilled water [18–20]. However, these models enable only qualitative evaluation, and cannot be applied to multi-pulse irradiation at high repetition rates. While the experimental measurement technique was developed half a decade ago [8,9], it is rather surprising there exists no analytical procedures to evaluate the nonlinear absorptivity only a year ago.

Very recently simulation model to evaluate the nonlinear absorptivity at high accuracy has been developed [21]. While the nonlinear absorptivity can be evaluated very precisely with an uncertainty of ±3% by the simulation model, some tedious numerical simulation is needed.

In this paper, the evaluation model is further extended to derive a simplified non-dimensional equation which is applicable to any glass materials under any experimental conditions including pulse energy, pulse repetition rate, translation speed, numerical aperture of focusing lens and so on.

2. Theoretical evaluation of W_{ab}

2.1. Simulation model for precise evaluation of W_{ab}

The simulation model is briefly outlined, since it is detailed elsewhere [21]. When a line heat source with continuous heat delivery of *w(z)* appears at $x=y=0$ in a region of $0<z<l$ in an infinite solid moving at a constant speed of v along x-axis as shown in Fig. 1, the steady temperature at (x,y,z) is given by

$$T(x,y,z)=\frac{1}{4\pi K}\int_0^l \frac{w(z')}{s}\exp\left\{-\frac{v}{2\alpha}(x+s)\right\}dz', \tag{1}$$

where $s^2=x^2+y^2+(z-z')^2$, $w(z)$ is distribution of absorbed laser power, K is thermal conductivity and α is thermal diffusivity given by $K/c\rho$ (c=specific heat and ρ=density). In this model, $w(z)$ and l are determined by fitting the isotherm to the experimental modification structure. $w(z)$ is determined by fitting the isotherm of the maximum cycle temperature under the condition of $dT/dx=0$ to the contour of the experimental modification structure. It is assumed that $w(z)$ is given in a simple form of $w(z)=az^m+b$ (a and b are constant) in order to reduce the number of unknown parameters in the fitting process.

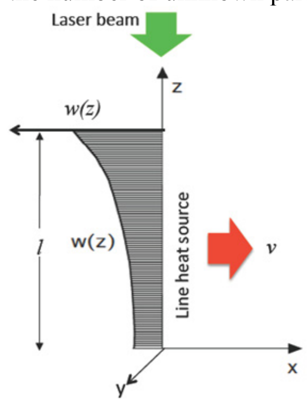

Fig. 1. Moving line heat source model with continuous heat delivery for simulating $w(z)$, l and W_{ab}.

Exact temperature distribution is available by such a simple line heat source model given by Eq.(1), because the temperature at locations apart from the heat source is spatially and temporally averaged. The intensity distribution *w(z)* and its extent *l* are determined by fitting of the simulated isotherm to the experimental modified structure. The average absorbed laser power W_{ab} is given by

$$W_{ab} = \int_0^l w(z)dz \tag{2}$$

2.2. Characterization of modified structure

In most commercially available glass, the modified region by ultrashort laser pulses exhibits dual-structure consisting of inner and outer regions, as shown in Fig. 2(a).

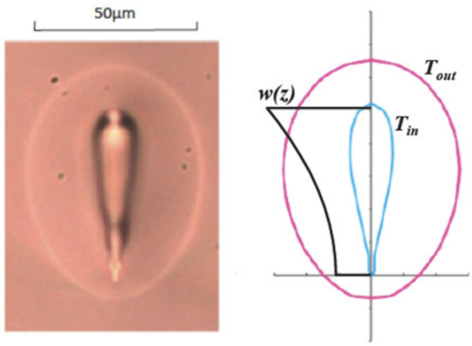

Fig. 2. (a) cross-section of internally modified D263; (b) thereby *w(z)* and *l*, assuming *m*=2

The concentration of the network modifiers and formers is reported to change rather steeply, since the diffusivity of the network modifiers is higher than that of network former at high temperatures [22]. We believe that the contour of the inner structure is visible, because concentration of the network formers and modifiers changes steeply near the contour of the inner structure. In general, however, the element distribution caused by diffusion process is not steep, and can be affected not only by temperature field but also by heating time. Actually our simulation results indicate that the characteristic temperature of the inner structure is not constant but ranges in 2800˚C~3600˚C depending on the experimental conditions. In addition, fused silica shows no inner structure, since no network modifiers are contained. These results suggest that the inner structure is not appropriate for fitting.

In overlap welding of glass plates of D263 by ultrashort laser pulses, it is observed that the glass plates are joined in the outer region [21], indicating that the contour of the outer region is actually melted. In other words, the outer region reaches the forming temperature defined by the viscosity $\eta=10^4$ *dPas*. Bovatsek *et al*. also showed that the joining is accomplished in the outer region [23]. So the isotherm of the forming temperature (T_F=1051˚*C* in D263 [24]) is fitted to the well-defined outer structure in the present study.

The simulated isotherms are plotted in Fig. 2(b) by fitting the isotherm of 1051˚C to the outer structure. The isothermal line of the inner structure is estimated to be approximately 3600˚C in this condition.

While our simulation can exactly determine the absorbed laser intensity distribution *w(z)*, extent of the absorbed region *l* and the nonlinear absorptivity A, tedious numerical calculation is needed. The purpose of the present study is to derive a simple non-dimensional equation, which can universally determine the nonlinear absorptivity of different glass materials with different thermal properties and experimental conditions without numerical simulation.

2.3. Derivation of simple non-dimensional equation for W_{ab}

In this section, a simple non-dimensional equation is derived to evaluate W_{ab} from the experimental cross-sectional area of the molten region S_{out}.

The non-dimensional equation cannot be derived from the equation of the line heat source model given by Eq.(1) because of the term of *w(z)*. So the line heat source model is approximated by the point heat source model [25], which corresponds to the case *l* in Eq.(1) is infinitesimally small given by

$$T(x,y,z)=\frac{W_{ab}}{4\pi rK}\exp\left\{-\frac{v}{2\alpha}(x+s)\right\}dz', \tag{3}$$

At the isothermal line of $T=T_F$, Eq.(3) is written in a non-dimensional form (z=0)

$$\Phi=\frac{1}{\sqrt{X^2+Y^2}}\exp\left(-X-\sqrt{X^2+Y^2}\right), \tag{4}$$

where $\Phi=vW_{ab}/8\pi T_\Phi K\alpha$, $X=vx/2\alpha$ and $Y=vy/2\alpha$.

The maximum cycle temperature at Y in the sample moving at a velocity of v is reached at the location of $X=X_m$ which is given by $d\Phi/dX$=0. In Fig. 3, X_m is plotted vs. Y by closed circles, which is approximated by

$$X_m \approx (AY)^B \tag{5}$$

where A=0.83 and B=1.818. Substituting Eq.(5) into Eq.(4), the following non-dimensional equation is derived.

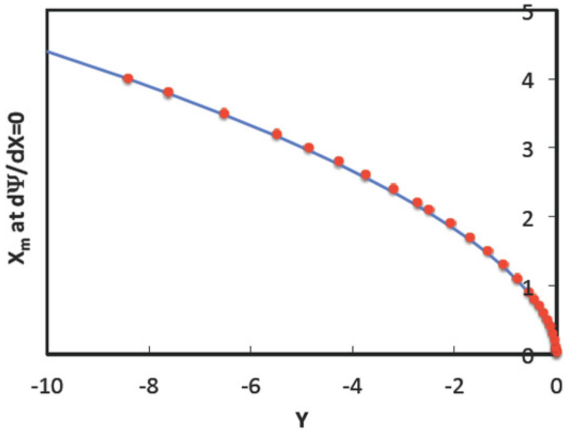

Fig. 3. Location of X_m at which maximum cycle temperature is reached plotted vs. Y in Eq.(4). Solid line is approximated by $X_m=(AY)^B$, where A=0.83 and B=1.818

$$\Psi = \frac{\sqrt{X_m^2 + Y^2}}{Y} \exp\left(X_m + \sqrt{X_m^2 + Y^2}\right) \tag{6}$$

where $\Psi = W_{ab}/4\pi K T_F y$. It is assumed that the cross-sectional area is S_{out}, y is given by $y=(S_{out}/\pi)^{1/2}$.

The nonlinear absorptivity can be determined at an accuracy as high as ±3% by the simulation model described in Sec. 2.1 [21]. So the simulated results can be used as the standard to evaluate the accuracy of the non-dimensional equation to be derived in this study.

3. Evaluation of W_{ab} at different materials

3.1. Relationship between W_{ab} and S_{out}

Ultrashort laser pulses of 10*ps* duration with a wavelength of 1,064 *nm* are tightly focused into the bulk glass by an objective lens for microscope with NA0.55 when the glass sample is translated in *x*-direction (Fig. 1).

In this experiment, three glasses, fused silica, D263 and Foturan, having wide variety of thermal properties are used. Fig. 4 shows thermal diffusivity and forming temperature of these glasses. Fused silica is characterized by high thermal diffusivity and high forming temperature. D263 and Foturan are characterized by low thermal diffusivity and low forming temperature, respectively. The bulk glass was internally melted at wide variety of conditions including pulse energy, pulse repetition rate, translation speed and NA of lens. The cross-sectional area of the outer region S_{out} was determined assuming the modified region is ellipse. The average absorbed laser power W_{ab} was evaluated by fitting the isotherm of T_F to the contour of the outer structure as is described in Sec 2.1.

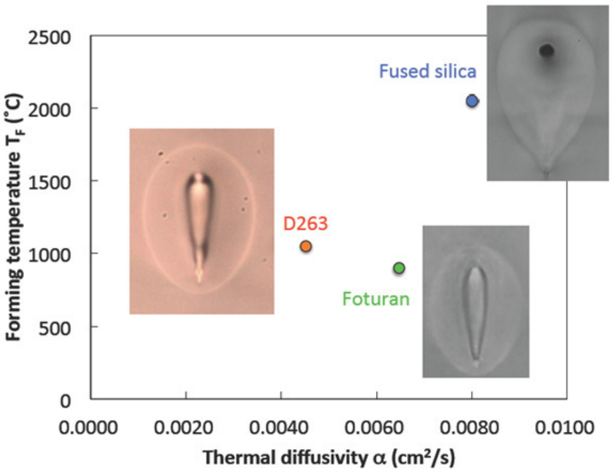

Fig. 4. Forming temperature and thermal diffusivity of D263, Foturan and fused silica used in the present study

Fig. 5 shows W_{ab} plotted vs. S_{out} for fused silica and D263 and Foturan. As S_{out} increases, W_{ab} increases monotonically, and the increasing rate becomes larger as the translation speed increases. The increasing rate of W_{ab} of fused silica is considerably larger than Foturan and D263 because of its higher forming temperature and thermal diffusivity.

3.2. Non-dimensional equation

In Fig. 6, relationship between Ψ and Y given by Eq.(6) is plotted by a solid line, which is slightly curved concavely as Y increases. The data in Fig. 5 are also re-plotted in Fig. 6. It is noted that the data points fall on a line with narrow scattering, showing slight deviation from Eq.(6). The data points can be approximated by a straight line given by

$$\Psi = aY + b \tag{7}$$

where a=1.1 and b=0.9.

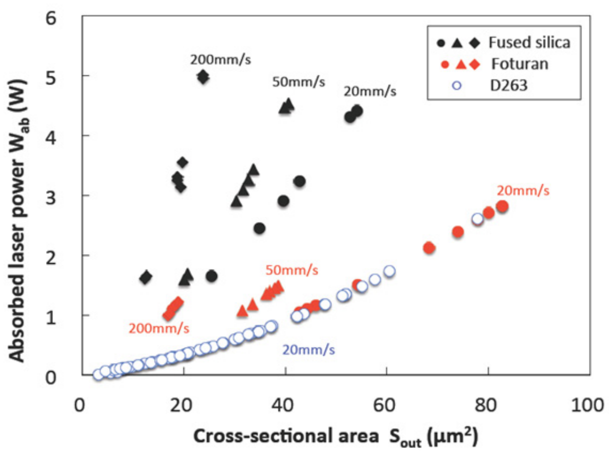

Fig. 5. Relationship between S_{out} and W_{ab} for fused silica, D263 and Foturan at different translation speeds, pulse energies, pulse repetition rates and lens NA

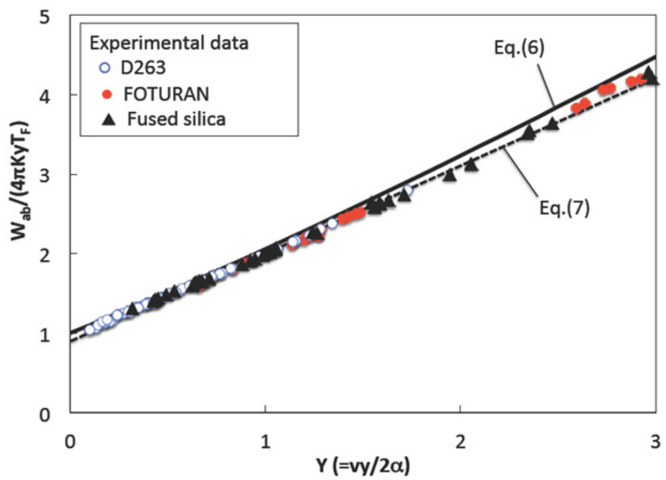

Fig. 6. Non-dimensional relationship between Y and Φ. Solid line and dotted line correspond to Eq.(6) and Eq.(7), respectively

The deviation of Eq.(7) from Eq.(6) is attributed to approximating the line heat source by the point heat source. Fig. 7(a) shows the aspect ratio *V/H* (*V*=height and *H*=width) of the cross section within the isothermal line of T_F, and Fig. 7(b) shows SL/SP (SL and SP are cross-sectional area of line heat source and point heat source, respectively) plotted vs. V/H under wide variety of experimental conditions. The aspect ratio V/H ranges in a region of 1.3~2.6, which corresponds to S_L/S_P=1.06~1.16 and Ψ_L/Ψ_P=1.03~1.08 (Ψ_L *and* Ψ_P are value of Ψ corresponding to line heat source and point hat source, respectively). If Ψ_L/Ψ_P=1.03~1.08 is taken into consideration, the deviation caused by the point heat approximation is larger at smaller Y, and hence it is understandable that the experimental data fall on rather straight line in Fig. 6.

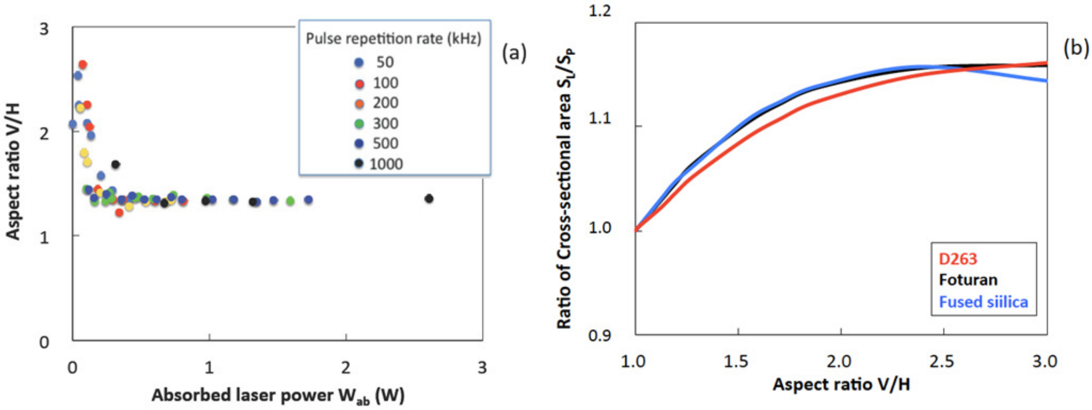

Fig. 7. (a) aspect ratio in D263 at different pulse energies and pulse repetition rates; (b) ratio SL/SP plotted vs. aspect ratio

4. Conclusions

Non-dimensional equation has been derived to correlate between S_{out} and W_{ab} in internal melting of glass by ultrashort laser pulses at high pulse repetition rates. It is shown that the evaluated nonlinear absorptivity has an excellent correlation with the results obtained by previously reported simulation model, based on a series of experiment using three glasses including D263, Foturan and fused silica having different thermal properties in wide a variety of experimental conditions.

Acknowledgments

This work was partially supported by Erlangen Graduate School in Advanced Optical Technologies (SAOT).

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